D02 · Capacity & Long-Range Planning

03 · Capacity capex decisions & business case

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

The structured process for justifying, sizing, and timing capital investments in production capacity — new lines, debottlenecking projects, equipment upgrades, greenfield plants. It converts the capacity gap identified in long-range planning (cap_02) into a financial case that competes for capital against every other corporate priority, with NPV, IRR, payback, and explicit treatment of uncertainty through scenarios and real options.

2 WHY IT MATTERS

Capacity capex is the most consequential decision Supply Chain owns — and the most frequently delegated to Finance by default. **The decision is irreversible in practice**: once a line is built, it generates depreciation, fixed cost absorption pressure, and a constituency that will resist mothballing it. The cost of over-building shows up for a decade; the cost of under-building shows up as missed revenue that is invisible to the income statement. The standard tools — NPV, IRR, payback — are necessary but insufficient: they implicitly assume a single demand scenario, which is exactly the assumption that breaks in capacity decisions. **Real options analysis** exists for this reason **(Trigeorgis, 1996 — foundational text)**: the right to expand later, to defer the decision, or to abandon mid-project has measurable value that traditional DCF ignores.

3 HOW IT IS DONE

1. **Anchor the case in the long-range capacity gap, not in available capex budget.** If the gap analysis (cap_02) says nothing has to be built, no business case rescues a "nice-to-have" project. If the gap is real, the case starts from a quantified shortfall — not from a vendor proposal.
2. **Build the NPV model with three demand scenarios — not one weighted average.** Run the case for pessimistic, base, and optimistic demand paths. A case that only works in the optimistic scenario is a case that does not work; the discipline is to size the investment to the case where the project still clears the hurdle rate in the base.
3. **Size the investment in modular phases when uncertainty is high.** One Phase-1 line with a Phase-2 expansion option (qualified site, reserved infrastructure, pre-engineered foundation) is structurally superior to a single full-capacity build when demand is unproven. The modular case forfeits some unit-cost economy in exchange for retained optionality.
4. **Quantify the option value, not just the NPV.** The right to defer, the right to expand, and the right to abandon mid-project all have positive value when uncertainty is high. A Black-Scholes-style calculation is overkill in most operational settings; a structured "what is this option worth in scenarios A/B/C" analysis is sufficient and defensible.
5. **Hold the case against the corporate hurdle rate, plus an execution-risk premium.** Greenfield projects in regulated environments warrant a hurdle rate premium that brownfield debottlenecking does not. A case that clears the corporate WACC but ignores execution risk is a case that will surprise the board in year two.
6. **Define explicit triggers for go/no-go decisions at each stage gate.** Capex projects do not fail at approval; they fail at the third unbudgeted scope change. Define before the first dollar is spent: under what observed conditions does the project pause, pivot, or kill — and who decides.

IN PRACTICE

Run the case across three demand scenarios

Build the business case on low/base/high demand, not the planning point forecast. The spread between scenarios is the information the decision actually needs; a single-scenario case hides exactly the uncertainty that should drive the sizing and the structure of the investment.

Size modularly when the spread is wide

When the gap between low and high scenarios is large, prefer a modular investment that can be scaled as demand resolves over a monolithic one. Modularity buys the option to expand without committing the full capital up front — the most direct way to convert irreducible demand uncertainty into manageable, staged risk.

Quantify the option value of waiting or expanding

Make the value of deferral and staged expansion explicit in the case, alongside the base NPV. Leadership cannot weigh flexibility it cannot see; pricing the option turns "build now vs wait" from a gut call into a quantified comparison.

Define go/no-go triggers before committing

Agree the demand or market signals that will trigger each stage of the investment before the first dollar is spent. Pre-defined triggers convert a staged plan into a discipline; without them, the optimistic scenario quietly becomes the case by default and the modularity is never actually exercised.

WHAT YOU WOULD USE

CAPEX MODELING / INVESTMENT ANALYSIS

Anaplan / SAP (IBP + EPM) / Oracle — enterprise tier-1 — Scenario-based capital and capacity modeling integrated with the financial plan and the long-range capacity view. **(Anaplan, SAP, Oracle — vendor)** The choice when the capex case must reconcile with the corporate financial plan and the operational capacity analysis.

Board / EPM platforms — scenario business cases — Connected-planning and EPM tools strong at multi-scenario investment cases and modular sizing. **(Board, EPM vendors — vendor)** A strong fit where the capex decision is driven primarily by the finance and scenario modeling.

Real-options / decision-analysis specialists — Dedicated real-options and decision-analysis tools for valuing flexibility, deferral, and staged commitment. **(specialized vendors — vendor)** The vertical option when the option value is large enough to justify rigorous quantification beyond standard DCF.

For operations that don't need a platform — A three-scenario DCF in a workbook with a modular-vs-monolithic comparison and explicit go/no-go triggers. The decisive discipline is refusing to run the case on a single forecast — and pricing the option to wait — which a spreadsheet handles as well as any platform for a single decision.

4 SO WHAT

Capacity capex is a decision under uncertainty, not under forecast. Run the case across three demand scenarios, size the investment modularly when the spread between scenarios is wide, quantify the option value of waiting or expanding later, and define the go/no-go triggers before the first dollar is committed. Supply Chain owns the discipline that prevents the optimistic scenario from becoming the case by default.

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